

ANALYZING SEA LEVEL RISE ADAPTATION & MITIGATION STRATEGIES IN
LOUISIANA AND THE NETHERLANDS

By

Jonah McDowell

AN UNDERGRADUATE THESIS

Presented to the Faculty of

The Environmental Studies Program at the University of Nebraska-Lincoln

In Partial Fulfillment of Requirements

For the Degree of Bachelor of Arts

Major: Environmental Studies

With the emphasis of: Policy, Advocacy, and Social Justice

Under the supervision of Dr. Ursula Kreitmair and Dr. Martha Durr

Lincoln, Nebraska

May 2023

ANALYZING SEA LEVEL RISE ADAPTATION & MITIGATION STRATEGIES IN LOUISIANA AND THE NETHERLANDS

Jonah McDowell, B.A.

University of Nebraska, 2023

Thesis Advisor: Dr. Ursula Kreitmair

Thesis Reader: Dr. Martha Durr

Abstract

Sea level rise is an increasingly dangerous threat. As a byproduct of climate change, this is an issue that has the potential to impact countless people, animals, and ecosystems across the globe. As sea level rise continues to worsen, it is imperative that coastal communities develop their own strategies and plans to combat this problem. This study focuses on two areas, the state of Louisiana and the country of the Netherlands, and studies how they have responded to this problem. These locations were chosen because sea level rise is proportionally more severe at these two locations compared to other areas of the world.

Two questions guided the research. The first question was “How do Louisiana and the Netherlands compare in their efforts to address rising sea levels?” One of the main focuses of this study was to review the strategies in use in both locations, and then see how they are similar or different. The second question was “What can we learn from these approaches to sea level preparedness?” By studying these areas, the strategies they used could potentially be examples for other communities to follow. A separate goal of this study was to create an all-encompassing sea level rise plan after reviewing the strategies of these two areas.

The research yielded the conclusion that they use very different methods to deal with the rising sea levels. In Louisiana, they preferred to use mitigation strategies. This entails the construction of levees along the coastline, structural protections for buildings, and habitat and ecosystem restoration projects. In the Netherlands, they use adaptation measures. Adaptation requires integrating strategies into society, as opposed to building large protections like Louisiana.

There are a lot of takeaways from a study like this, but one of the most important ones is the fact that each coastal location will have their own unique strategy for dealing with sea level rise. Too many factors determine which strategies are most effective to implement. The most important thing is that coastal communities are prepared to deal with this issue.

Introduction

Considerable sea level rise is expected in the future, with devastating impacts on coastal communities. As soon as 2045, more than 300,000 homes will be at risk for chronic inundation (Union of Concerned Scientists, 2018). Despite these statistics, many coastal communities are unprepared to deal with this approaching problem. This thesis will systematically assess the flood preparedness efforts taken by the state of Louisiana and the European country of the Netherlands to identify lessons learned and provide recommendations for other locations. These areas were chosen due to the prevalence of sea level rise on their coastline, and because they are of different cultures. By studying how they both respond to the issue, conclusions can be made about which ones are most beneficial to use.

Sea-level rise is slowly but surely surrounding homes and places of business, until areas are no longer habitable. What are cities doing to address *that*? The very real possibility that thousands of people will be uprooted from their homes as they sink into the ocean, what are answers to that question? Does it require a comprehensive plan? How does culture fit in? Finance? Are these areas prioritizing adaptation or mitigation? Maybe the answer is a mixture of both. The purpose of this research isn't to try and find the "right answer" because there isn't one. Civilization hasn't dealt with something like this before, so there is no right and wrong, it's simply about values. What is prioritized? The scope of the issue and solution are things to be discovered within the research.

Research Plan

For clarification purposes, 'efforts' means the specific policies and/or community initiatives that these two areas have implemented, or at least talked about. For the purposes of this project, sea level rise is the slow, incremental, encroaching of the ocean on the land that people live on and work on. This study seeks to systematically assess how two at-risk locations (the Netherlands and Louisiana) prepare for expected sea-level rises, with the intention to provide recommendations for other communities seeking to adapt to this particular climate change risk. The assessment will consider various location characteristics that may make adoption and success of strategies more or less likely. Things like culture, geography, and financial constraints will all be taken into consideration when providing a recommendation. The two areas of interest were chosen for distinct reasons. Both of these two areas, though one is a state and one is its own independent nation, are similar in their relationship with water. Louisiana and the Netherlands both constantly deal with the threat of rising water levels, and have dealt with these crises on numerous occasions. What this study aims to look deeper at is the factors that contribute to their level of preparedness.

Project Significance

This topic is important for a number of reasons. First, it could be one of the effects of climate change that will have the greatest impact on people's perception of climate change. The switch hasn't really flipped in the vast majority of people's minds that climate change is

happening and its effects will be catastrophic. Once people see the streets of New Orleans or Amsterdam underwater, never to reappear again, then maybe the issue will finally be taken seriously on the global scale. Secondly, this is important to me because when these communities are flooded, we will see a refugee crisis unlike anything we've ever seen. In fact, current projections claim that by the end of this century, 2 billion people across the world could find themselves as climate refugees due to rising sea levels (Cornell, 2017). There have been plenty of refugee crises in history, and plenty of climate-related ones in the past years. But none in the United States, or any other major country. That is another aspect that will hopefully wake everyone up once and for all about climate change. Odds are you will know someone affected, or know someone who knows someone who's affected, and it will force everyone to care about the issue. But again, the question is where will these people go? Is our country equipped to deal with that sort of crisis? That isn't the focus of this thesis, but it's adjacent to it. This research should take place because it is almost certain that these issues will come to fruition. There are only a handful of locations that have specific climate adaptation plans, so it is important to analyze the ones that exist. The study provides insights in an area with significant real life applications for an issue with considerable risks.

Background Information/Literature Review

Rising Sea Levels

As our planet continues to warm, it causes the polar ice caps to melt at an accelerated pace. Massive ice shelves break off and fall into the ocean, eventually melting and mixing with the rest of the water. Isolated incidents may not seem like a big deal, but when this occurs on a global scale, as it is now, it has the potential for catastrophic consequences. One of, if not the, most harmful product of this phenomenon is sea level rise. Human civilization across the globe calls the beaches and shores of the ocean home, and they are under threat. Unfortunately, the global trends are going in the wrong direction. Since 1880, the global sea level has risen by an average of 8-9 inches (Climate, 2022). Recently we had our highest recorded figure, as the global sea level in 2021 was 3.8 inches higher than it was in 1993 (Climate, 2022). These numbers are staggering, and point toward a bleak future if the nations of the world refuse to act on the issue of climate change. In fact, according to recent studies, no matter how much global emissions are lowered in the coming years, it is almost certain that average global sea levels will rise by 15-30cm by the year 2050 (IPCC, 2021). This number is substantial, and could increase even more in the second half of the century depending on our ability to successfully lower emissions on a drastic scale. However, it is important to not lose hope. Many initiatives are being done to address this problem of sea level rise, some of which will be later discussed in this paper.

A Lack of Preparedness for Sea-Level Rises

Sea level rise preparedness can take on two forms: adaptation and mitigation. Adaptation can be described as "the process of adjusting to the current and future effects of climate change", while mitigation is "making the impacts of climate change less severe by preventing or reducing

[its effects]” (European Environmental Agency, 2022). Cities and communities dealing with sea level rise will likely deploy a combination of these two strategies. There are a number of strategies agreed upon across the world as being effective strategies to address this problem. “Flood barriers to protect critical infrastructure include levees, dikes, and seawalls” (Indiana University, 2023). New York City is utilizing this strategy, as they are currently “building a \$335 million flood wall in Manhattan” (SeaLevelRise.org, 2023). An important thing to keep in mind is that though these efforts are critical and necessary, they will come with a large price tag. That is a mitigation strategy; an adaptation strategy would be to “purchase coastal land that is damaged or prone to damage and use it for conservation” (Indiana University, 2023). It is important to have these ecosystems along the coast, as they “act as natural buffers for coastal areas during rainstorms and hurricanes. They absorb precipitation and storm surge waters” (Clean Ocean Action, 2023). In addition, using beaches and natural infrastructure to our advantage is a proven way to address the problem. “Beaches and dunes can act like a natural wall that reduces the impact of storm surges.... Natural structures such as barrier islands, oyster and coral reefs, mangroves, seagrass, and salt marshes can work in unison with built infrastructure, such as seawalls, to absorb storm surges” (SeaLevelRise.org, 2023). Other strategies include implementing more stormwater pumps in coastal areas, as “pumps can speed up the process of getting water off the streets by vacuuming up the flood water and releasing it back into the sea” (SeaLevelRise.org, 2023).

The Need for Evidence-Based Recommendations

Because the issue of sea level rise due to climate change has only relatively recently become more urgent, it is still unclear as to which methods of adaptation and mitigation are the most effective. This is precisely why studying the approaches of New Orleans and the Netherlands is important, and will help us fill in some of those gaps.

Though the Results section will go into greater detail regarding the similarities and differences between the two areas, it's important to provide a brief overview here as well. Louisiana, as most Americans know, constantly deals with the threat of tropical storms and hurricanes. Because of their geographical location, they must prepare for flooding because of storms, in addition to their rising coast levels. As a result, this area has invested a lot of resources into mitigating the effects of these floods and storms. Governments at the state and local level in Louisiana have made this issue a priority, as the plans included later in the text show. The Netherlands, however, are able to approach the issue differently. Because of their location on the North Sea, the tropical storms and hurricanes that plague Louisiana are not a concern. They have to focus on the sea rising and flooding their streets, and their infrastructure and plans reflect that. Like Louisiana, this is a problem at the forefront of the minds of government officials and the general population. Because it is of such great urgency, it highlights the importance that their plans and strategies are studied. Their efforts could help other locations better prepare for their own flooding events.

Methods

Research/Data Collection Plans

As the information in the above paragraphs show, this project will heavily consist of qualitative information. Because the goal of this project is to compare and contrast the responses of the Netherlands and Louisiana, the information will revolve around plans, policies, adaptations, and mitigations. Social research will be a big contributor as well. This research is about “investigating and seeking answers to the social questions that we and others ask around our social world” (Walter, 2013). There are many social questions that need to be answered as to how these communities will change and adapt to this threat of rising sea level. Both communities have addressed many issues, as the text below details. But there is still a long way to go, and there will always be people who disagree. The way these communities deal with pushback will show a great deal about their culture, and about their cohesiveness.

In addition to social research, this project will follow the form of a historical design as well. The purpose of this form of research design is to “collect, verify, and synthesize evidence from the past to establish facts... It uses... official records, reports...” (USC, 2006). Though the word ‘historical’ may have the connotation of having taken place in the distant past, that isn’t the case. All of the information cited in this paper is historical because it documents things that these two areas have already implemented. By using data from governmental sites and other sources with direct access to these issues, it is ensured that the most accurate and relevant information is being used.

Available Documents

It was difficult at first to find relevant information for this project. Due to the recentness and presentness of the issue of rising sea levels, there was not much in the way of published academic articles on sea level rise in Louisiana and the Netherlands. Despite using key words and phrases like “sea level rise”, “mitigation and adaptation strategies”, “Louisiana”, “the Netherlands”, “governmental plans”, and “coastal sea level rise”, articles directly relating to the focus of this study didn’t show up. Because of this, and after discussion with Dr. Gosselin, the research was shifted to online sources. Using similar keywords and phrases, searching via Google provided much more relevant and up to date information as it pertained to the subject area. From there, official adaptation and resilience plans for these two areas, along with other relevant sources, were obtained to determine which strategies the people of Louisiana and the Netherlands were implementing.

The information will be obtained in different ways depending on the location. For Louisiana, publicized plans and recommendations are very accessible and easy to reference. The Resilient New Orleans Plan, utilized in the Results section, is a great example of this type of source. As shown in the previous sections, multiple areas and groups of people in Louisiana have created adaptation and mitigation strategies in the form of polished, effective plans. For the Netherlands, that isn’t the case. Though the city of Rotterdam has created the *Rotterdam Adaptation Strategy*, most of the information obtained about their situation was provided in the

form of articles. Some of these article sources include Dutch Review, the New York Times, Dutch Water Sector, and de Urbanisten. This, of course, makes it more difficult to verify that the information is factual. To combat this, when finding an article that might seem relevant, it is imperative to review the rest of the site first. This will hopefully provide a clearer picture as to the sort of information that the source produces, and a decision can be made based on whether or not the source can be trusted. Of course, it isn't ideal that the information from the two locations are coming in different forms. However, it is necessary in this instance, and still allows for an effective method of finding information.

Reviewing the contingency plans and mitigation/adaptation strategies that each location has implemented is important for discerning differences in approaches. In this way, comparisons can be drawn. The main goal of this project is to create a plan recommendation for a hypothetical location that is dealing with the issue of rising sea levels based on the strategies of the Netherlands and Louisiana. It is imperative, then, to carefully review what each location has done and determine if that strategy is worth adding to the plan recommendation. Other approaches, like interviewing the residents of the areas for their input, wouldn't yield the same information as the actual plan implementations themselves. That is why this information gathering method is so crucial to the success of the project.

Information will only be used if it is from an official source, or from an article that is widely known and respected as factual. If information is found from a source where it is unclear whether or not it can be trusted, then that information will not be used. Under the guidance of social research and historical design, the data collection process will be methodical and use factual information. Most often, that information will be the various policies and plans these areas have implemented.

Results

Louisiana -- Protecting Against Water

To start, let's look at Louisiana. Specifically, New Orleans. One thing the city has done has been to create a *Resilient New Orleans Plan*. This plan places a huge emphasis on prioritizing and protecting the poorer communities from the effects of sea level rise (Al, 2018). This is important for a number of reasons, most of them obvious. First, these communities are made up of citizens that don't have the means of evacuating and "surviving" a flood on their own. If it destroys their home and their belongings, they have nowhere to turn to get any sort of repayment for that. This plan keeps these people in mind, and provides resources for them in the event that their homes are lost or impacted by sea level rise. Secondly, sort of similar to the first reason, we know that politicians and governments rarely pass ordinances meant to specifically benefit the less fortunate. Because of this, having a plan set forth by the community to make sure they are cared for in the event of an emergency is very good to see.

Evacuation Plan, Culture, & Restoration

Along those same lines, the city established a City-Assisted-Evacuation system for its residents in the event of a natural disaster or major storm (Landrieu, 2017). This is an adaptation

measure, but it's very helpful to the people who need a way to get out of town. As the city's "Climate Action for a Resilient New Orleans" document says, they believe that "to be successful (in addressing sea level rise), we must also create a culture of awareness and action about climate change and resilience" (Landrieu, 2017). Now that sounds good, but how will they achieve that? One way is by connecting climate action to the rich culture of New Orleans. They hope that by making it seem like a community and culture issue instead of a scientific one, they will be able to educate their citizens and inspire them to make sustainable changes too. They also believe that local leaders are the way forward in the climate crisis, not the federal level (Landrieu, 2017). Overall, the plan wants to prepare for the impacts of climate change by advancing coastal protection and restoration, investing in comprehensive and innovative urban water management, and incentivizing property owners to invest in risk management (Landrieu, 2017). Part of that preparation includes working to implement home elevations and building code improvements, as well as transitioning to more renewable forms of energy, as they will be more reliable in the event of a crisis (Landrieu, 2017). These aren't the only actions taking place, however. Through the *Coastal Protection and Restoration Authority Strategic Plan*, ecosystem restoration strategies will be implemented via partnerships with federal, state, and local government entities (CPRA, 2022). These strategies will include high priority projects to protect, restore, enhance, and create coastal wetland systems in Louisiana. Additionally, the plan hopes to practice flood protection strategies, projects, and activities (like building miles of levees), to reduce economic losses from storm-based flooding (CRPA, 2022).



Map of Louisiana Sea Level Rise

Mitigation Strategies & Infrastructure

To finish looking at Louisiana, we will dive into *Louisiana's Comprehensive Master Plan for a Sustainable Coast*. A huge amount of resources have been dedicated to mitigation strategies, better preparing the coastlines for future floods. Since 2007, the state has completed 282 miles of levee improvements (Edwards, 2017), and the plan builds off that. 124 projects are

planned to build/maintain over 800 square miles of land, and would reduce expected flood damage by \$8.3 billion annually over the next 50 years (Edwards, 2017). Additionally, \$19 billion has been allocated for structural protections, as well as making flood-proofing more than 1,400 structures and elevating more than 22,400 buildings a top priority (Edwards, 2017). Clearly, the state has decided that protecting their coasts from the rising sea levels is something they want to tackle head-on. While they made mitigation efforts, they have also addressed the natural environment itself. The state built a long distance sediment pipeline that has transferred nearly 10 million cubic yards of Mississippi River sediment to support projects designed to better protect the communities of south New Orleans (Edwards, 2017). The state has also created over 1,000 acres of marsh and 11,000 linear feet of restored ridge in Bayou Dupont, showing that the state is making a point to restore their coastline in order to protect both their communities, and the natural habitats and wildlife. These efforts show a very well-rounded approach to dealing with the issue.

Community Input

While everything listed above is very important, they weren't the main components of the plan. The majority of the plan revolved around community input and collaboration. Instance after instance in the plan includes some communication with the people of the community, or experts in specific fields. The plan places an emphasis on coastal communities, and got input from focus groups representing these communities in the forms of fisheries, navigation, energy, and industry (Edwards, 2017). These instances of communication took place in informal open-house and/or small group discussions to help facilitate ideas. These smaller groups allowed more people to let their ideas and thoughts be heard, whereas large group discussions might alienate or discourage people from speaking their mind as freely. The plan also created a community focus group to help guide the development and implementation of the master plan, while also fostering partnerships with tribal communities and local community advocates (Edwards, 2017). As the plan was released in 2017, the creators of the plan hosted four official public hearings across the state to receive feedback and comments from the public. Over 800 people participated in the hearings, and were allowed to enter their comments electronically, in writing, or face-to-face with a plan representative at the hearings (Edwards, 2017). This is an amazing display of respect for the input of Louisiana residents, and it's something all public policy makers and environmentalists should strive to achieve. As you can see, there are numerous examples of the plan placing value on input from the public, which is a very progressive and good thing to see. On top of that, it's important to hear from the experts as well. The plan coordinators are working with floodplain managers, hazard mitigation specialists, other state agencies, and NGO's to make sure the goals they set are attainable and worthwhile. Additionally, they are working alongside the U.S. Army Corps of Engineers on flood risk reduction and large-scale restoration planning (Edwards, 2017). In other words, the plan involves collaboration between federal, state, and local partnerships.

Both New Orleans and the state of Louisiana have done a great job showing the potential that a well-thought out climate plan can have. They addressed key issues like restoration

strategies, mitigation and adaptation implementation, protection of the natural habitats of Louisiana, prioritizing the safety of their residents, and getting input from them at the same time. As the water levels on the coast of Louisiana are expected to rise by between 1 ½ to 2 feet by 2050 (Schleifstein, 2022), the plans already in place are very positive for the people living there. It was a very well thought-out and well executed plan that touched on every key issue, while at the same time keeping the immediate needs of the public at the forefront.

The Netherlands -- Living With Water

Now we will turn our attention to the Dutch. They deal with the same problems as the citizens of Louisiana, but deal with it in a slightly different manner.

Culture & Community Input

In the Netherlands, there is an acceptance of the water. It's thought of as simply a part of their life. They aren't trying to figure out ways to stop the water from coming, but instead focus on how they can structure their way of life around the water. In other words, they are attempting to live in harmony with the water. This is perfectly encapsulated by Rotterdam's climate chief when he said "It's not just a bunch of dikes and dams, but a way of life" (Kimmelman, 2017). This way of thinking is shown in the *Rotterdam Adaptation Strategy*. Addressing a number of issues, flood adaptation and protection is a main focal point. Under the plan, "the 'sponge function' of the city will be restored by implementing measures to capture and store rainwater where it falls and to delay drainage." Through an "intensive participation process, various community stakeholders... all put forth good ideas that were included in the design for the Watersquare" (de Urbanisten, 2013). For them, the water provides an opportunity to build around. The officials in the Netherlands view environmental and social resilience as going hand-in-hand, through efforts to improve neighborhoods, spreading equity, and taming water during catastrophes.

Infrastructure

For example, the Dutch have built garages, plazas, and parks that serve as both an aspect of daily life and as enormous reservoirs for when the water comes (Kimmelman, 2017). Another example of this philosophy is a newly built rowing course utilized by the public and professionals alike, while also serving as a floodwater collection area during emergencies (Kimmelman, 2017). More recently, the city of Rotterdam in 2013 built the first-ever Watersquare of Benthemplein, a recreation area that can hold up to 1.7 million liters of water in the event of flooding. When the flooding ends, the water retained in the Watersquare will flow into a nearby canal and the area will be ready for recreational use again (DutchReview, 2021). As these examples show, building higher walls and levees is not the route taken over there. They believe in working with the water instead of around it. Further examples represent these ideals through the usage of a national GPS-guided app to inform residents exactly how far below sea level they are at any given time. Another is the requirement of children to earn diplomas while swimming in clothes and shoes before they are allowed to use public pools unrestricted. This is cited as a safety issue in the event of a flood, and "a basic part of our culture, like riding a bike"

(Kimmelman, 2017). You see the differences in this way of thinking compared to Louisiana. Both are addressing the issue at hand, but while Louisiana is fighting against the water, the Netherlands seem to be working in tandem with it. As the mayor of Rotterdam said, “If the water comes in... we can evacuate maybe 15 out of 100 people. So evacuation isn’t an option... we must learn to live with water” (Kimmelman, 2017). In accordance with that way of thinking, the country has installed plazas with fountains, gardens, and basketball courts in underserved neighborhoods that can act as retention ponds in the event of a flood (Kimmelman, 2017). Similar to Louisiana, they are making sure the poorer communities aren’t being left to fend for themselves.

Mitigation Strategies

While they largely rely on living with water, they have implemented some flood mitigation strategies as well. Most notably, the creation of the Maeslantkering. This massive structure serves as a sea gate for the port of Rotterdam, protecting both the town itself and the economic importance of the port (Kimmelman, 2017). Additionally, the creation of the *Room for the River* plan. This plan aims to manage higher water levels in four major rivers in the Netherlands by lowering the levels of floodplains, creating water buffers, relocating levees, increasing the depth of side channels, and starting the construction of flood bypasses (Netherlands, 2019). The four targeted rivers are the Rhine, the Meuse, the Waal, and the IJssel. At over 30 locations along these rivers, there have been measures to give the rivers space to flood safely (Netherlands, 2019). This has been a very involved plan, with 19 partners from provinces, municipalities, regional water authorities, and the Rijkswaterstaat (group responsible for the maintenance of Dutch infrastructure facilities) all working together to get this done (Netherlands, 2019).

The Netherlands have been able to put climate action high on people’s list of priorities because they have shown how improving public spaces can translate into urban renewal and the economic value of investing in resilience. Instead of simply building dams and levees, they have incorporated flood mitigation strategies into everyday life, and have seen economic successes because of it.



Maeslantkering

Comparing the Two Cases

For this project, the research questions revolved around the issue of sea level rise on the coasts of Louisiana and the Netherlands. How do the two areas differ in their adaptation and mitigation strategies? How are they similar? After diving into those questions, the objective was to distinguish their similarities and differences, and then create an independent plan based on the strategies that the two areas have used.

As described above, Louisiana has invested a lot of resources and time to develop plans for various communities across the state. Through the *Resilient New Orleans Plan*, *Coastal Protection and Restoration Authority Strategic Plan*, and *Louisiana's Comprehensive Master Plan for a Sustainable Coast*, the area has made it clear the policies and strategies they will and have put into place. The Netherlands have used a different approach. Though not publishing thorough plans like Louisiana has, they have implemented a handful of adaptation strategies with an underlying theme of living with the water. The table below will outline these strategies in greater detail.

To obtain the information included in the table below, the process laid out in the Methods section was followed. This process ensures that the information is relevant and factual.

Table 1: Comparison of Louisiana and Dutch Approaches

	Louisiana	The Netherlands
Published plans for addressing the issue?	Yes; <i>Resilient New Orleans Plan</i> , <i>Coastal Protection and Restoration Authority Strategic Plan</i> , and <i>Louisiana's Comprehensive Master Plan for a Sustainable Coast</i>	Yes; <i>Rotterdam Adaptation Strategy</i>
Community input?	Yes, one of the main components of <i>Louisiana's Comprehensive Master Plan for a Sustainable Coast</i> was community input. Constituents voiced their opinions through focus groups, open houses, public hearings, and electronically.	Yes; the <i>Rotterdam Adaptation Strategy</i> included an “intensive participation process, various community stakeholders... all put forth good ideas that were included in the design for the Watersquare”
Mitigation strategies?	Yes; flood protection (building miles of levees), projects aimed at building/maintaining over 800 square miles of land, funds allocated for structural protections and flood-proofing buildings across the state, built a sediment pipeline to better support projects that protect southern New Orleans communities	Yes, but not the main focus; the Maeslantkering is a massive structure that protects the city and port of Rotterdam, the <i>Room for the River</i> plan lowers the levels of floodplains, creates water buffers, relocates levees, etc.
Adaptation strategies?	Yes, but not the main focus; New Orleans established a City-Assisted-Evacuation system for residents in the event of a disaster or major storm	Yes; <i>Rotterdam Adaptation Strategy</i> focusing on flood adaptation and protection, garages/plazas/parks that serve as an aspect of daily life and as enormous reservoirs in the event of flooding/rising sea levels, Watersquare of Benthemplein in Rotterdam, national GPS-guided app to inform residents exactly how far below sea level they are at any given time, requirement of children to earn diplomas while swimming in clothes and shoes before they are allowed to use public pools unrestricted
Reliance on adaptation or mitigation?	Mitigation; levels and coastal restoration strategies a main point in their plans	Adaptation; they really prioritize “living with the water”, as evidenced by their dual-purpose retention ponds
Evacuation Plans	The <i>Resilient New Orleans Plan</i> established a “city-assisted-evacuation system” for its residents	As the mayor of Rotterdam said, “If the water comes in... we can evacuate maybe 15 out of 100 people. So evacuation isn’t an option... we must learn to live with water” (Kimmelman, 2017).
Infrastructure	Home elevations and building code improvements, hundreds of miles of levees, allocating billions for structural protections, building a sediment pipeline to transport millions of cubic yards of river sediment to protect coastal communities, created/restored ecosystems to act as natural infrastructure	Dual purpose garages/plazas/parks that also act as reservoirs, Watersquare (recreation area that can hold up 1.7 million liters of water), Maeslantkering (sea gate protecting the port of Rotterdam)

Discussion

Let's review what the results of the study tell us. Evidenced by the three published plans that the state of Louisiana has created, they are taking a very regimented, structured approach. All their goals and strategies are carefully laid out for all to see, and they refer to those documents when implementing adaptation and/or mitigation strategies. As their plans show, the state heavily relies on mitigation strategies as opposed to adaptation. Implementing things such as levees, reinforcing structural protections and flood-proofing coastal buildings show their tendency to build up their communities to fight against the water. The Netherlands has approached the issue in a very different way. Evidenced by their building of dual-purpose community areas like plazas, parks, and garages that also serve as retention ponds, the people of the Netherlands prioritize living with the water instead of against it. While they have mitigation strategies, like the Maeslantkering, they have invested a great deal more in integrating rising water levels into their everyday lives.

One of the objectives of this study was to create a recommendation after reviewing the strategies of these two areas. The determination that can be made from this study is that the recommendation really depends on the specific area. There is no blend of adaptation and mitigation strategies that creates the perfect sea-level rise protection. Infrastructure, history, culture, and geography all play a role in determining the best route forward. Louisiana frequently deals with tropical storms, has a very large coastline, and is below sea-level, requiring them to utilize mitigation strategies. Areas with similar attributes, like Florida or areas of Mexico along the Gulf of Mexico would benefit from following the example that Louisiana has set. The Netherlands are one of Europe's oldest nations, and their infrastructure and towns reflect that. These communities include homes and cities right on top of the water, presenting an interesting situation. They don't have ample space to build levees like Louisiana does, which is why it makes all the more sense that they use mainly adaptation strategies instead of mitigation. Parts of Belgium and Germany, who are below sea-level like the Netherlands and often live right on top of the water, would be smart to implement similar adaptation strategies that the Netherlands have used.

Summary & Conclusions

Overview

This study was conducted to address one of the most pressing issues of our generation. Sea-level rise is a byproduct of global warming, and its effects will be felt by millions now and in the future. Almost 75% of the global population lives within 30 miles of the ocean (World Economic Forum, 2022), so this is a very prevalent issue. To get a closer look at how certain communities are addressing this problem, the study focused on the efforts of Louisiana and the Netherlands. These two areas are dealing with the same issue of sea-level rise, but approached their strategies very differently. This study aimed to compare the two areas, and discern what could be learned from their differing strategies.

Findings

The study provided insight into the strategies used by both Louisiana and the Netherlands. Louisiana approached the issue through the lens of mitigation. By implementing levees and structural protections, their efforts focused on fortifying their coastline from rising waters. Additionally, the state has created a handful of published plans, highlighting their desire to have a strong structure as they move forward in addressing this growing threat. On the other hand, the Netherlands dealt with the issue in a different way. They focus on living with the water, instead of fighting against it like Louisiana does. An example of this mindset is their requirement for children to pass a swim test in full clothes and shoes before being able to use local public pools. This test prepares them so that in the event of sudden flooding due to rising sea-levels, they have experience swimming in their normal clothes.

Further Study

This study can and should be expanded upon, in large part due to the importance of this issue. That being said, there are a few areas that this study wasn't able to address that would be useful to examine further. One of these areas would be to interview people from each area and ask them about their strategies. Discussion provides insight and reasoning that articles and plans often don't, and these insights would be very useful when trying to get a better understanding of why these areas make the decisions they do. Another thing that could be beneficial is looking closely at the funds each area has allocated to this issue. Preliminary research was unsuccessful in finding this information, but it is likely available somewhere. Knowing the budget that each area has allocated toward addressing this problem would provide great insight into how much they are prioritizing it. Additionally, it would provide a monetary baseline for other communities. Whether they are looking to use measures like either Louisiana or the Netherlands has, knowing how much money they spent on their strategies would be very helpful for communities looking to adopt their own strategies to know. This is another area where an interview would help, as an interviewer could ask questions about their funding. Along those same lines, a question left unanswered is where these areas received the funding to implement the strategies they have. Did it come from taxpayers or private donations? This distinction would also be very important to know, as it would offer a more all-encompassing look into the process of implementing these strategies. Related to that, a shortcoming of this study is that it doesn't much discuss the process of how these decisions and strategies were made, it simply talks about the end result. By approaching the study from a more chronological point of view, the findings could be used as a blueprint for other coastal communities. How does the process begin? What or who determines if a project will be implemented? How long does it take to create these structures? Are elected officials involved? If so, how? Questions like these are extremely important to obtaining a more holistic viewpoint of how this process occurs, and definitely needs to be addressed in order to understand the process better.

References

Clean Ocean Action. Clean Ocean Action: Be the Solution to Ocean Pollution. (n.d.). Retrieved from <https://cleanoceanaction.org/be-the-solution-to-ocean-pollution>

Climate action for a Resilient New Orleans: Mayor's Office of Resilience and Sustainability. (2017, July). Retrieved October 20, 2022, from <https://nola.gov/climate-action/>

Coastal Protection and Restoration Authority. (2017, June 2). Retrieved October 21, 2022, from http://coastal.la.gov/wp-content/uploads/2017/04/2017-Coastal-Master-Plan_Web-Book_CFinal-with-Effective-Date-06092017.pdf

Coastal Protection and Restoration Authority. (2022). Retrieved October 21, 2022, from <https://coastal.la.gov/wp-content/uploads/2022/06/Strategic-Plan-Update-2023-2024-through-2027-2028-FINAL.pdf>

Cornell University. "Rising seas could result in 2 billion refugees by 2100." ScienceDaily. www.sciencedaily.com/releases/2017/06/170626105746.htm (accessed December 4, 2022).

Garcia-Fuentes, L. (2021, July 06). Rising sea levels in the Netherlands: The Dutch battle against flooding. Retrieved from <https://dutchreview.com/culture/society/rising-sea-levels-in-the-netherlands/#:~:text=One%20of%20the%20most%20impressive,North%20Sea%20into%20the%20land.>

Indiana University. (n.d.). *Sea level rise*. Environmental Resilience Institute. Retrieved from <https://eri.iu.edu/erit/strategies/sea-level-rise.html>

Kimmelman, M., & Haner, J. (2017, June 15). The Dutch have solutions to rising seas. the world is watching. Retrieved October 20, 2022, from <https://www.nytimes.com/interactive/2017/06/15/world/europe/climate-change-rotterdam.html>

Kopp, Robert. "IPCC Climate Report: Profound Changes Are Underway in Earth's Oceans and Ice." China Dialogue Ocean, February 10, 2022. [https://chinadialogueocean.net/en/climate/18217-ipcc-climate-report-profound-changes-are-underway-in-earths-oceans-and-ice/amp/?gclid=Cj0KCQiA-JacBhC0ARIsAIxybyOEnK3oVF2k7lbjLCrZomzYrIma9HM4G-mR8VBHHudLINWe02iwW_waAsJXEALw_wcB.](https://chinadialogueocean.net/en/climate/18217-ipcc-climate-report-profound-changes-are-underway-in-earths-oceans-and-ice/amp/?gclid=Cj0KCQiA-JacBhC0ARIsAIxybyOEnK3oVF2k7lbjLCrZomzYrIma9HM4G-mR8VBHHudLINWe02iwW_waAsJXEALw_wcB)

Lindsey, Rebecca. "Climate Change: Global Sea Level." NOAA Climate.gov. Accessed December 6, 2022. <https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level>.

Only a sixth of the world's coastlines remain in their natural state. World Economic Forum. (2022, February 15). Retrieved from <https://www.weforum.org/agenda/2022/02/ecologically-intact-coastlines-rare-study/>

Organizing your Social Sciences Research Paper: Types of research designs. (n.d.). Retrieved November 16, 2022, from <https://libguides.usc.edu/writingguide/researchdesigns>

Room for the river programme. (2019, April 15). Retrieved October 20, 2022, from <https://www.dutchwatersector.com/news/room-for-the-river-programme>

Rotterdam Climate Proof. (2013). Rotterdam adaptation strategy. Retrieved from <https://www.urbanisten.nl/work/rotterdam-adaptation-strategy#:~:text=The%20Rotterdam%20adaptation%20strategy%20sets,the%20sea%20and%20the%20rivers.>

"Sea Level Rise." National Geographic Society. Accessed December 9, 2022. <https://education.nationalgeographic.org/resource/sea-level-rise>.

SeaLevelRise.org. (n.d.). *Sea level rise solutions*. Sea Level Rise. Retrieved from <https://sealevelrise.org/solutions/>

Schleifstein, Mark. "Seas Could Rise 2 Feet on Louisiana Coast by 2050, 4 Feet by 2100, Federal Officials Say." NOLA.com, February 17, 2022. https://www.nola.com/news/environment/seas-could-rise-2-feet-on-louisiana-coast-by-2050-4-feet-by-2100-federal/article_9d7d7868-8f6d-11ec-8228-ffaebd3d9f8c.html#:~:text=30%2C%202021.,15%2C%20by%20NOAA%20and%20NASA.

Underwater. (2018, June 18). Retrieved from https://www.ucsusa.org/resources/underwater?utm_source=googlegrants&utm_medium=search&utm_campaign=climate&gclid=CjwKCAiAlp2fBhBPEiwA2Q10D-6dC6hxyCO1YCP9BVtcvRtnvx8yDvyJPdGKLs7fa3TnRIg0TxWSZxoCLhIQAvD_BwE

Walter, M. (2013). The Nature of Social Science Research. Retrieved November 16, 2022.

What is the difference between adaptation and mitigation? European Environment Agency. (2022, July 28). Retrieved from [https://www.eea.europa.eu/help/faq/what-is-the-difference-between#:~:text=In%20essence%2C%20adaptation%20can%20be,\(GHG\)%20into%20the%20atmosphere.](https://www.eea.europa.eu/help/faq/what-is-the-difference-between#:~:text=In%20essence%2C%20adaptation%20can%20be,(GHG)%20into%20the%20atmosphere.)